Section A: Multiple Choice Questions

A particle is in simple harmonic motion of amplitude A and total energy E. At one instant its kinetic energy is 3E/4. Its displacement at that instant is

(A) $A/\sqrt{2}$
(B) $A/2$
(C) $A/\sqrt{(3/2)}$
(D) $A/\sqrt{3}$
(E) A

A 1-kg object attached to a spring with force constant 25.0 N/m oscillates on a horizontal frictionless surface. At time $t = 0$ the object is released from rest at x = +3.0 cm (that is the spring is stretched by 3.0 cm). The correct equation of motion for the object coordinate in SI units is:

(A) $x(t) = (0.030\ \text{m})\cdot\cos(0.50\ t+\pi/3)$
(B) $x(t) = -\ (0.030\ \text{m})\cdot\cos(5.0\ t+\pi/2)$
(C) $x(t) = (0.030\ \text{m})\cdot\cos(5\ t-\pi/3)$
(D) $x(t) = (0.030\ \text{m})\cdot\cos(5.0\ t)$
(E) $x(t) = -(0.030\ \text{m})\cdot\cos(5.0\pi\ t+\pi/6)$

A wave traveling on a string is described by the expression $y = 0.005 \sin(20x - \omega t)$, where y is in meters, and x and t are in meters and seconds, respectively. If the speed of the wave is 20.0m/s, what is its frequency?

(A) 31.9 Hz
(B) 63.7 Hz
(C) 0.64 Hz
(D) 400 Hz
(E) 0.16 Hz

Transverse waves with a speed of 5.4 m/s are to be produced in a taut string. A 5.00-m long string with a mass of 375 g is used. What is the required tension?

(A) 0.2 N
(B) 0.4N
(C) 2.2 N
(D) 0.6 N
(E) 1.5 N

The superposition of two waves,

$$y_1 = (2\times10^{-8}\ \text{m})\sin\left[\pi\left(\frac{x}{2\ \text{m}} - \frac{170}{\text{s}}t\right)\right]$$

and

$$y_2 = (2\times10^{-8}\ \text{m})\sin\left[\pi\left(\frac{x}{2\ \text{m}} - \frac{170}{\text{s}}t - \frac{1}{2}\right)\right],$$

results in a wave with a frequency of

(A) 85 Hz.
(B) 170 Hz.
(C) 85π Hz.
(D) 340 Hz.
(E) 170π Hz.

Two harmonic waves are described by

$$y_1 = (3.00\ \text{cm})\sin\left[\pi\left(\frac{2.00}{\text{m}}x + \frac{3.00}{\text{s}}t\right)\right]$$

$$y_2 = (3.00\ \text{cm})\sin\left[\pi\left(\frac{2.00}{\text{m}}x - \frac{3.00}{\text{s}}t\right)\right]$$

What is the magnitude of the displacement (in cm) of this wave at x = 3 cm and t = 5 sec?

(A) 6.0
(B) 1.50
(C) 3.00
(D) 1.13
(E) 0

The path difference between two waves is 4 m. If the wavelength of the waves emitted by the two sources is 5m, what is the phase difference (in radians)?

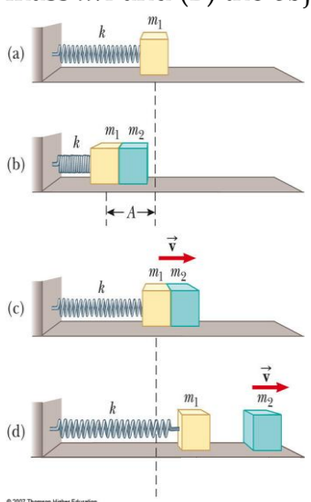
(A) 90
(B) 288
(C) 1.57
(D) 5.03
(E) 7.85

Standing waves are produced by the interference of two traveling sinusoidal waves, each of frequency 100 Hz. The distance from the second antinode to the fifth antinode is 60 cm. The wavelength of each of the two original waves is:

(A) 50 cm
(B) 40 cm
(C) 30 cm
(D) 20 cm
(E) 15 cm

Section B: Long-Answer Questions

9. An object of mass m_1=9.00 kg is in equilibrium, connected to a light spring of constant k=100 N/m that is fastened to a wall as shown in the diagram (a) below. A second object, m_2=7.00 kg, is slowly pushed up against m_1, compressing the spring by the amount 0.200 m (see diagram (b)). The system is then released and both objects start moving to the right on a frictionless surface. At time t=0 m_1 reaches the equilibrium, m_2 loses contact with m_1 and moves to the right with constant speed. Write the equations in SI units describing the position of (A) the object of mass m_1 and (B) the object of mass m_2 after the separation.



Energy $E = \frac{1}{2} k A^2$

at $x = 0$: $U = 0$ $E = \frac{1}{2}(m_1+m_2) v^2 =$

$\frac{1}{2}\, 100 \frac{N}{m} (0.200\,m)^2 = 2.00\,J$

$v^2 = \frac{2E}{m_1+m_2} = \frac{2 \times 2.00\,J}{16.00\,kg} = 0.25 \frac{m^2}{s^2}$

+3 $v = 0.5\,m/s$

$\boxed{\text{Object 2: } x_2(t) = vt = 0.5t\,m}$ +1

Object 1: $x_1(t) = A\cos(\omega t + \varphi)$ +2

$\omega = \sqrt{\frac{k}{m_1}} = \sqrt{\frac{100\,N/m}{9.00\,kg}} = \boxed{3.33\,rad/s}$ +1

$v = v_m = \omega A \rightarrow A = \frac{v_m}{\omega} = \frac{0.5\,m/s}{3.3\,rad/s} = \boxed{0.150\,m}$ +1

$t = 0$: $x = 0 \rightarrow \varphi = \frac{\pi}{2}$ or $\frac{3\pi}{2}$

$v > 0 \rightarrow v = -\omega A \sin\varphi > 0 \rightarrow \boxed{\varphi = \frac{3\pi}{2}}$ +1

$\boxed{x_1 = (0.150\,m)\cos\left(3.33t + \frac{3\pi}{2}\right) = (0.150\,m)\sin(3.33t)}$ +1

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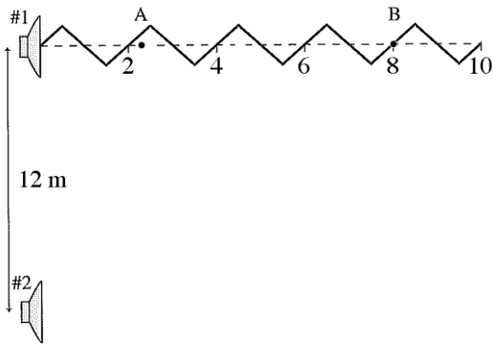
10. Two speakers 12 m apart emit identical sound waves with a wavelength of 2 m. The two waves are in phase as they leave the speakers. The distances from speaker 1, r_1, are shown in the diagram.



(a) Will the interference at point A located at r_1=2.20 m be ~~constructive or~~ destructive?

$r_1 = 2.20\ m \quad r_2 = \sqrt{2.20^2 + 12^2} = 12.2\ m$ (+1)

$\Delta r = r_2 - r_1 = 12.2 - 2.2 = 10.0\ m$

$\Delta r = 5\lambda \rightarrow$ $\boxed{\text{constructive}}$ (+1)

(b) To have destructive interference at point B, located at r_1 = 8 m, at what frequency (closest to the original frequency) should the speakers emit? Assume the speed of sound in air is 340 m/s.

Destructive: $\Delta r = (2n+1)\frac{\lambda}{2}$ (+1)

(+1) $\Delta r = r_2 - r_1 \quad r_1 = 8\ m, \quad r_2 = \sqrt{8^2 + 12^2} = 14.4\ m$ (+1)

$\Delta r = 14.4 - 8 = 6.42\ m$ (+1)

$\Delta r = (2n+1)\frac{\lambda}{2} = (2n+1)\frac{v}{2f}$ (+1)

$f = \frac{(2n+1)v}{2\Delta r} = \frac{(2n+1)(340\ m/s)}{2(6.42\ m)} = (2n+1)26.47\ Hz$ (+1)

Original frequency: $f = \frac{v}{\lambda} = \frac{340\ m/s}{2\ m} = 170\ Hz$

Find n for orig. freq.: $170\ Hz = (2n+1)26.47\ Hz$

$\downarrow n = 2.7$ (+1)

Closest integer $n \rightarrow n = 3$

$f = (2 \cdot 3 + 1)\ 26.47 = \boxed{185.29\ Hz}$ (+1)

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11. A standing wave pattern is observed in a thin wire with a length of 2.00 m. The wave function is $y = (0.003\text{ m}) \sin(2\pi x) \cos(50\pi t)$ where x is in meters and t is in seconds.

(a) How many nodes does this pattern exhibit?

$\lambda = \frac{2\pi}{k} = \frac{2\pi}{2\pi\, m^{-1}} = 1m.$ $\quad \lambda_n = \frac{2L}{n} = 1m$

$n = \frac{2L}{1m} = \frac{2 \cdot 2.00\,m}{1\,m} = 4 -$ number of loops $\rightarrow$ $\boxed{5 \text{ nodes}}$ 2

(b) What is the fundamental frequency of vibration of the wire?

$\omega = 50\pi \frac{rad}{s}$; $f_n = n\frac{v}{2L}$; $f_4 = \frac{\omega}{2\pi} = 25\,Hz$;

$f_1 = \frac{f_n}{n} = \frac{25}{4}\,Hz = \boxed{6.25\,Hz}$ 2

(c) If the original frequency is held constant and the tension in the wire is increased by a factor of 4, how many nodes are present in the new pattern?

$\lambda = \frac{v}{f}$; $v = \sqrt{\frac{T}{\mu}}$ 2

$T' = 4T \rightarrow v' = \sqrt{\frac{4T}{\mu}} = 2\sqrt{\frac{T}{\mu}} = 2v$

$\lambda' = \frac{2v}{f} = 2\lambda = \frac{2L}{N} \rightarrow 2m = \frac{4m}{N}$

number of loops $N = 2 \rightarrow$ $\boxed{3 \text{ nodes}}$

(d) Write two wave functions of the original waves that can form this standing wave.

$y(x,t) = 2A \sin kx \cos\omega t =$ +2

$= A\sin(kx - \omega t) + A\sin(kx + \omega t)$

$A = \frac{0.003\,m}{2} = 0.002\,m$

$k = 2\pi\, m^{-1}$ $\quad \omega = 50\pi\ rad/s$ +2

$$\boxed{\begin{aligned} y_1 &= (0.002m)\sin(2\pi x - 50\pi) \\ y_2 &= (0.002m)\sin(2\pi x + 50\pi) \end{aligned}}$$

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